(REF, RREF, NF)

Zero Row: All elements of the row are zero.
Non-Zero Row: At least one element of the row is non-zero.
Pivot: First non-zero element of each row considered from the left.
Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

ELEMENTARY ROW OPERATION:

- The interchange of the ith row and jth row; and it is denoted by $Ri \leftrightarrow Rj$.
  Example:

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 4 \\ 2 & 4 & 8 \\ 0 & 0 & 0 \end{pmatrix} \quad R_2 \leftrightarrow R_3$$

- The multiplication of every element of the ith row by a nonzero scalar k; and it is denoted by $Ri \longrightarrow kRi$.
  Example:

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} \sim \begin{pmatrix} 5 & 15 & 20 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix} \quad R_1 \longrightarrow 5R_1$$

- The addition of the elements of the ith row and k times the jth row; and it is denoted by $Ri \longrightarrow Ri + kRj$.

$$M = \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 2 & 4 & 8 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 3 & 4 \\ 0 & 0 & 0 \\ 5 & 13 & 20 \end{pmatrix} \quad R_3 \longrightarrow R_3 + 3R_1$$

Row Echelon Form (REF):

Step 1: All zero rows, if any, are at the bottom of the matrix.

Step 2: Each pivot in a row is to the right of the pivot in the preceding row.

N.B: All entries in a column below a pivot are zero.

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

REF

Procedure:

1. Interchange positions of zero rows (whenever they appear) so that they lie below all nonzero rows.
2. Selection of pivots.
3. Formation of zeroes below pivots (using pivots).
   Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ \boxed{0 \quad 0 \quad 0 \quad 0} \\ -4 & 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_4 \longleftrightarrow R_3$$

R_3:	-4	0	0	0
$4R_1$:	4	8	-4	0
---	--	--	--	--
$R_3 + 4R_1$:	0	8	-4	0

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 8 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \longrightarrow R_3 + 4R_1$$

$3R_3$:	0	24	-12	0
$8R_2$:	0	24	32	4
---	--	--	--	--
$3R_3 - 8R_2$:	0	0	-44	-4

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \longrightarrow 3R_3 - 8R_2$$

REF

Reduced Row Echelon Form (RREF):

Step 1: All zero rows, if any, are at the bottom of the matrix.

Step 2: Each pivot in a row is to the right of the pivot in the preceding row.

Step 3: Each pivot is equal to 1.

Step 4: Each pivot is the only nonzero entry in its column.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

RREF

Procedure:

1. Make each pivot as 1.
2. Make zeroes above each pivot by means of that pivot.
   Example:

$$\begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & -44 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ REF}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 0 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R_3 \longrightarrow \frac{R_3}{-44}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 3 & 0 & 3/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_1 \longrightarrow R_1 + R_3 \\ R_2 \longrightarrow R_2 - 4R_3 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 3 & 0 & 3/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} R_1 \longrightarrow R_1 + R_3 \\ R_2 \longrightarrow R_2 - 4R_3 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 0 & 1/11 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} R_2 \longrightarrow \frac{R_2}{3}$$

$$\sim \boxed{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix}} R_1 \longrightarrow R_1 - 2R_2$$

RREF

Normal Form (NF):

1. Consist of only ones and zeros.
2. Every row has a maximum of single one and rest are all zeros (there can be rows with all zeros).
3. Every column has a maximum of single one and rest are all zeros (there can be columns with all zeros).

Example:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\boxed{(I_3)} \qquad \boxed{(I_3 \mid 0)}$$

$$\left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \hline 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

$$\boxed{\left(\begin{array}{c} I_3 \\ \hline 0 \end{array}\right)} \qquad \boxed{\left(\begin{array}{c|c} I_3 & 0 \\ \hline 0 & 0 \end{array}\right)}$$

Normal Form of a matrix

$$(I_n) \qquad (I_n \mid 0) \qquad \left(\begin{array}{c} I_n \\ \hline 0 \end{array}\right) \qquad \left(\begin{array}{c|c} I_n & 0 \\ \hline 0 & 0 \end{array}\right)$$

To find Normal Form of a matrix, we need to do column operation,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/22 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} \boxed{\text{RREF}}$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/11 \\ 0 & 0 & 0 & 0 \end{pmatrix} C_4 \longrightarrow C_4 - \frac{1}{22} C_2$$

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} C_4 \longrightarrow C_4 - \frac{1}{11} C_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{c|c} I_3 & 0 \\ \hline 0 & 0 \end{array}\right) \boxed{\text{NF}}$$